

Python Web Developer Roadmap

Python Web Developer Complete Checklist

1 Python Fundamentals

Learn the basics of Python.

- ☐ Install Python and set up environment
- ☐ Variables and data types
- ☐ Numbers and strings
- ☐ Lists, tuples, sets, dictionaries
- ☐ Conditional statements (if, elif, else)
- ☐ Loops (for, while)
- ☐ Functions
- ☐ Lambda functions
- ☐ Modules and packages
- ☐ File handling
- ☐ Error handling (try, except)
- ☐ Object-Oriented Programming (OOP)
- ☐ Virtual environments (venv)

Web Fundamentals

Understand how the web works.

- ☐ Basics of HTML
- ☐ Basics of CSS
- ☐ Basics of JavaScript
- ☐ How browsers work
- ☐ HTTP / HTTPS protocol
- ☐ Request vs Response
- ☐ GET / POST / PUT / DELETE methods
- ☐ Status codes (200, 404, 500)

Version Control

Learn version control using Git.

- ☐ Install Git
- ☐ `git init`
- ☐ `git add`
- ☐ `git commit`
- ☐ `git push`
- ☐ `git pull`
- ☐ Branches
- ☐ Merge conflicts
- ☐ Use GitHub

Databases

Learn SQL databases.

- ☐ Install PostgreSQL or MySQL
- ☐ Database basics
- ☐ Tables and columns
- ☐ Primary and foreign keys
- ☐ SQL queries (SELECT, INSERT, UPDATE, DELETE)
- ☐ JOIN queries
- ☐ Indexes
- ☐ Database relationships
- ☐ Use SQLite for small apps

Python Web Framework

Learn a framework.

Flask

- ☐ Install Flask
- ☐ Routes
- ☐ Templates (Jinja2)
- ☐ Forms
- ☐ Database integration
- ☐ REST API with Flask

Django

- ☐ Install Django
- ☐ Django project structure
- ☐ Models
- ☐ Views
- ☐ Templates
- ☐ Django ORM
- ☐ Authentication system
- ☐ Django Admin panel

FastAPI

- ☐ Install FastAPI
- ☐ Create API endpoints
- ☐ Request validation
- ☐ Response models
- ☐ Async APIs

APIs

Learn to build and test APIs.

- ☐ Understand REST APIs
- ☐ JSON format
- ☐ CRUD operations
- ☐ API authentication
- ☐ API documentation
- ☐ Test APIs with Postman

Testing

Test your applications.

- ☐ Unit testing
- ☐ Install PyTest
- ☐ Write test cases
- ☐ Test API endpoints

Deployment & DevOps

Learn how to run apps online.

- ☐ Linux basics
- ☐ Environment variables
- ☐ Install Docker
- ☐ Containerize your app
- ☐ Deploy on Amazon Web Services
- ☐ Deploy on DigitalOcean

Advanced Backend Topics

Important for professional developers.

- ☐ Authentication (JWT)
- ☐ Authorization
- ☐ WebSockets
- ☐ Background tasks
- ☐ Caching with Redis
- ☐ Task queues with Celery
- ☐ Security best practices

Projects

Build real projects.

- ☐ Todo App
- ☐ Blog Website
- ☐ REST API service
- ☐ Chat application
- ☐ E-commerce backend
- ☐ Authentication system
- ☐ Social media app

Portfolio

Show your work.

- ☐ Upload projects to GitHub
- ☐ Write README documentation
- ☐ Deploy projects online
- ☐ Create portfolio website

Job Preparation

Get ready for developer jobs.

- ☐ Practice coding challenges
- ☐ Prepare backend interview questions
- ☐ Build 5–10 strong projects
- ☐ Apply on LinkedIn
- ☐ Apply on Indeed